

Eleventh International Olympiad, 1969

1966/1.

Prove that there are infinitely many natural numbers a with the following property: the number $z = n4 + a$ is not prime for any natural number n .

1969/2.

Let $a_1, a_2, \dots, a_n$ be real constants, x a real variable, and

$$f(x) = \cos(a_1 + x) + \frac{1}{2}\cos(a_2 + x) + \frac{1}{4}\cos(a_3 + x) + \dots + \frac{1}{2^{n-1}}\cos(a_n + x). \quad (1)$$

Given that $f(x_1) = f(x_2) = 0$, prove that $x_2 - x_1 = m\pi$ for some integer m .

1969/3

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1969/4

For each value of $k = 1, 2, 3, 4, 5$, find necessary and sufficient conditions on the number $a > 0$ so that there exists a tetrahedron with k edges of length a , and the remaining $6 - k$ edges of length 1.

1969/5

A semicircular arc γ is drawn on AB as diameter. C is a point on γ other than A and B , and D is the foot of the perpendicular from C to AB . We consider three circles, $\gamma_1, \gamma_2, \gamma_3$, all tangent to the line AB . Of these, γ_1 is inscribed in $\triangle ABC$, while γ_2 and γ_3 are both tangent to CD and to γ , one on each side of CD . Prove that γ_1, γ_2 , and γ_3 have a second tangent in common.

1969/6

Given $n > 4$ points in the plane such that no three are collinear, prove that there are at least

$$\binom{n-3}{2}$$

convex quadrilaterals whose vertices are four of the given points.

1969/7

Prove that for all real numbers $x_1, x_2, y_1, y_2, z_1, z_2$, with $x_1 > 0, x_2 > 0, x_1y_1 - z_1^2 > 0, x_2y_2 - z_2^2 > 0$, the inequality

$$8(x_1 + x_2)(y_1 + y_2) - (z_1 + z_2)^2 \leq \frac{1}{x_1y_1 - z_1^2} + \frac{1}{x_2y_2 - z_2^2}$$

is satisfied. Give necessary and sufficient conditions for equality.